

MAT 5173 — Weekly Schedule

Algebra I · Fall 2026 · UT San Antonio

Texts

Primary text

- **Knapp** — Knapp, *Basic Algebra*

Secondary reading

- **Rosebrock** — Rosebrock, *Visual Group Theory*
- **Beardon** — Beardon, *Algebra and Geometry*
- **CLO** — Cox, Little & O’Shea, *Ideals, Varieties, and Algorithms*

Supplementary reading

- **Shafarevich** — Shafarevich, *Basic Algebraic Geometry 1*
- **Needham** — Needham, *Visual Complex Analysis, 25th Anniversary Edition*

Handouts & papers

- **Smith** — Smith, Kahanpaa, Kekalainen & Traves, *An Invitation to Algebraic Geometry*
- **Arnold–Rogness** — Arnold & Rogness, *Möbius Transformations Revealed*

Schedule

Date	Topic	Reading
Wed 19 Aug	Lecture — Course launch; the plane and its isometries	Rosebrock Ch. 1
Mon 24 Aug	Lecture — Reflections generate everything	Rosebrock Ch. 1
Wed 26 Aug	Lecture — From symmetry to the group axioms	Rosebrock Ch. 2; Knapp Ch. IV
Mon 31 Aug	Lecture — Cyclic and dihedral groups	Rosebrock Ch. 2
Wed 2 Sep	Lecture — Permutations and the sign homomorphism	Beardon §1.3-1.4; Knapp Ch. I (permutations)
Wed 9 Sep	Q&A (Due: Set 1)	
Mon 14 Sep	Lecture — Subgroups; Cayley graphs	Rosebrock Ch. 3
Wed 16 Sep	Lecture — Cosets and Lagrange	Beardon §12.2-12.3; Knapp Ch. IV
Mon 21 Sep	Lecture — Homomorphisms and quotient groups	Rosebrock Ch. 3; Beardon §12.11-12.12
Wed 23 Sep	Lecture — Group actions: orbits and stabilizers	Rosebrock Ch. 4; Knapp Ch. IV
Mon 28 Sep	Lecture — Orbit-stabilizer and counting	Rosebrock Ch. 4
Wed 30 Sep	Q&A (Due: Set 2)	

Date	Topic	Reading
Mon 5 Oct	Exam — Midterm Exam 1 — Sets 1 and 2	
Wed 7 Oct	Lecture — Symmetries of the polyhedra; finite rotation groups in space	Beardon §14.2-14.3
Wed 14 Oct	Lecture — Finite groups; frieze and wallpaper patterns	Rosebrock Ch. 7; Beardon §14.4
Mon 19 Oct	Lecture — Mobius transformations as a group action on the sphere	Beardon Ch. 13; Needham Ch. 3 §IV-VI (handout; a copy is on reserve); Mobius Transformations Revealed (2 min)
Wed 21 Oct	Lecture — Rings and polynomial rings $k[x]$, $k[x,y]$	Knapp §VIII.1; CLO §1.1; Beardon §2.3
Mon 26 Oct	Lecture — Integral domains and fields of fractions; polynomials in one variable	Knapp §VIII.2; CLO §1.5
Wed 28 Oct	Lecture — Ideals; prime and maximal ideals	Knapp §VIII.3; CLO §1.4
Mon 2 Nov	Q&A (Due: Set 3)	
Wed 4 Nov	Lecture — Affine space and affine varieties $V(S)$	CLO §1.2; Smith Ch. 1 (handout)
Mon 9 Nov	Lecture — $I(V)$ and the algebra-geometry dictionary	CLO §1.4
Wed 11 Nov	Lecture — The coordinate ring $k[x,y]/I(V)$ — the landing	CLO §1.4; Knapp §VIII.4; Reading guide, Segment B
Mon 16 Nov	Lecture — Why $\mathbb{C}$ and not $\mathbb{R}$: the Nullstellensatz, stated	CLO §1.4; Reading guide, Segment B
Wed 18 Nov	Q&A (Due: Set 4)	
Mon 23 Nov	Lecture — Plane curves — rational, cusp, node	Shafarevich §1.1
Mon 30 Nov	<i>Review</i> — Review — the exam pool	
Wed 2 Dec	Exam — Midterm Exam 2 — Sets 3 and 4	